

Periodic RVE: high-dimensional model

Displacement snapshots and effective responses
Fixed material and training domain

Projection-based state approximation

Affine: $\mathbf{u}_{\text{ref}} + \mathbf{V}_{\text{tra}} \mathbf{q}_{\text{tra}}$

Nonlinear: $\mathbf{u}_{\text{ref}} + \mathbf{V} \mathbf{q} + \bar{\mathbf{V}} \mathcal{N}(\mathbf{q})$

PANNs and comparison baselines

Regression: direct stress output

Free: unrestricted scalar energy

ICNN / ICKAN: constrained scalar energy

Hyperreduced-order models

HPROM: fixed ECM and equilibrium

HPROM-ANN: MAW-ECM and equilibrium

D-HPROM-ANN: direct stress cubature

Constituent law on selected elements

Direct constitutive evaluation

No microscopic mesh or equilibrium solve

Energy models: differentiate the potential

Regression: differentiate the stress map

Guarantees depend on the architecture

Illustrative structural deployment

Common macroscopic FE solver and stress/tangent interface

Same mesh, loading and stopping criteria